

1 INPUTS

Seed prompt p_0
(five fixed sections)

Mining set M
failures shown to
the optimizer

Validation set V
selects best round;
flips shown to optimizer

**Sealed
test set T**

seed prompt, M , V

2 BASELINE

Solver LLM scores
the seed prompt p_0
on M and V

Round 0 saved
(p_0, s_0, v_0)

Baseline test score
of p_0 on T
(measured, not used)

round-0 scores; start round 1

3 REFLECTIVE ROUND LOOP (rounds 1 to 3)

$t = 1$: no reflection yet

Reflect
what did the last
edit recover and
regress on M , V ?

**Optimizer LLM
writes a new
prompt p_t**

Solver LLM
scores p_t
on M and V

Save round t
always kept,
no gate

loop back for rounds 2 and 3

after round 3: all saved rounds

second and last use

4 SELECT AND REPORT

Pick best round t^*
by validation accuracy

**Evaluate shipped prompt
 p^* on T (final use)**

Final result
baseline vs. shipped,
recovered / regressed

measured only: never used to choose, edit, or stop